

Solar Pump Monitoring Controller

1. Introduction

The Solar Pump Monitoring Controller is designed to automate water pumping operations while ensuring battery protection, dry-run protection, and pump health monitoring. The system continuously monitors solar voltage, battery voltage, pump current, tank level, pump feedback, and dry-run conditions. Based on these inputs, the controller automatically controls pump operation and provides fault indications.

The project is implemented using STM32F103 as the primary controller and ESP32 as the IoT communication module. Sensor and Current System data are transmitted from STM32 to ESP32 through UART communication and uploaded to Firebase Realtime Database for cloud monitoring.

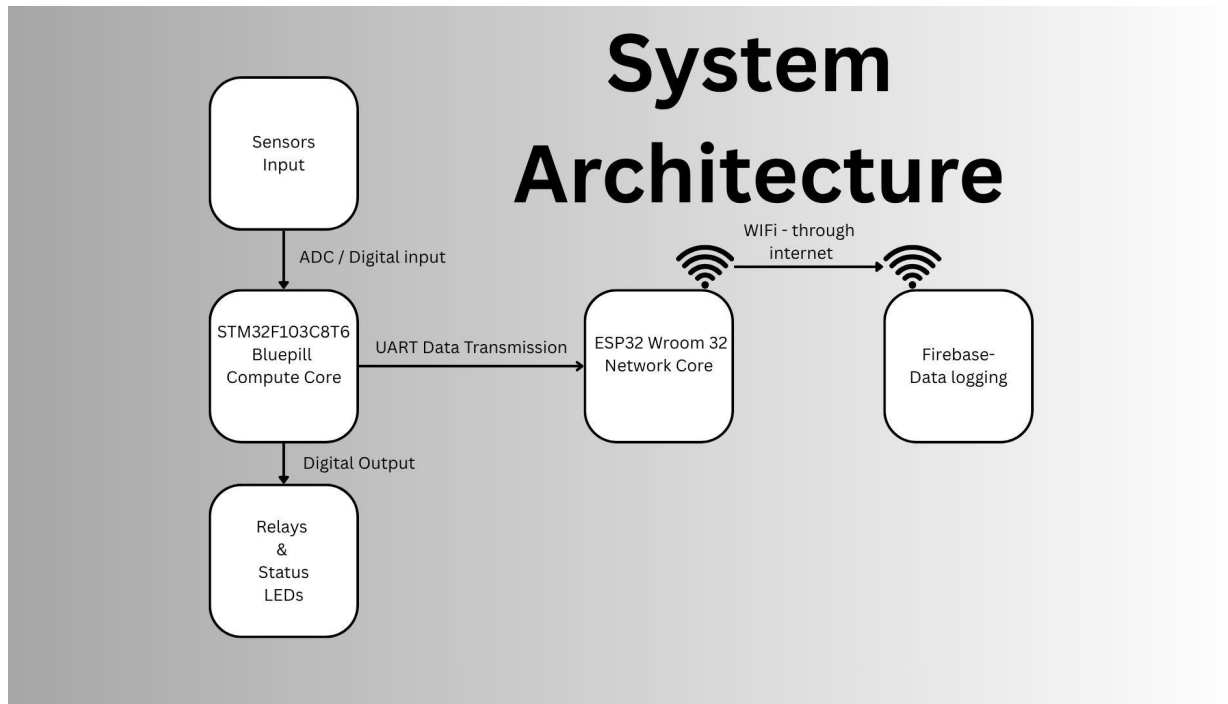
2. System Architecture

The system consists of two controllers:

STM32F103 (Main Controller)

Responsible for:

- Sensor data acquisition
- ADC measurements
- Pump control logic
- Battery protection logic
- Dry-run protection logic
- Alarm generation logic
- UART data transmission
- Status Leds logic
- Charging LED logic



ESP32 (IoT Controller)

Responsible for:

- Receiving UART data from STM32
- Parsing system parameters
- Connecting to Wi-Fi
- Uploading data to Firebase Realtime Database

3. Hardware Components Used

Controller Boards

- STM32F103C8T6 Blue Pill Development board
- ESP32 WROOM 32 Development board

Sensors

- Potentiometer for Solar Voltage Simulation
- Potentiometer for Battery Voltage Simulation
- Potentiometer for Pump Current Simulation
- Water Level Sensor for Tank Water level Simulation

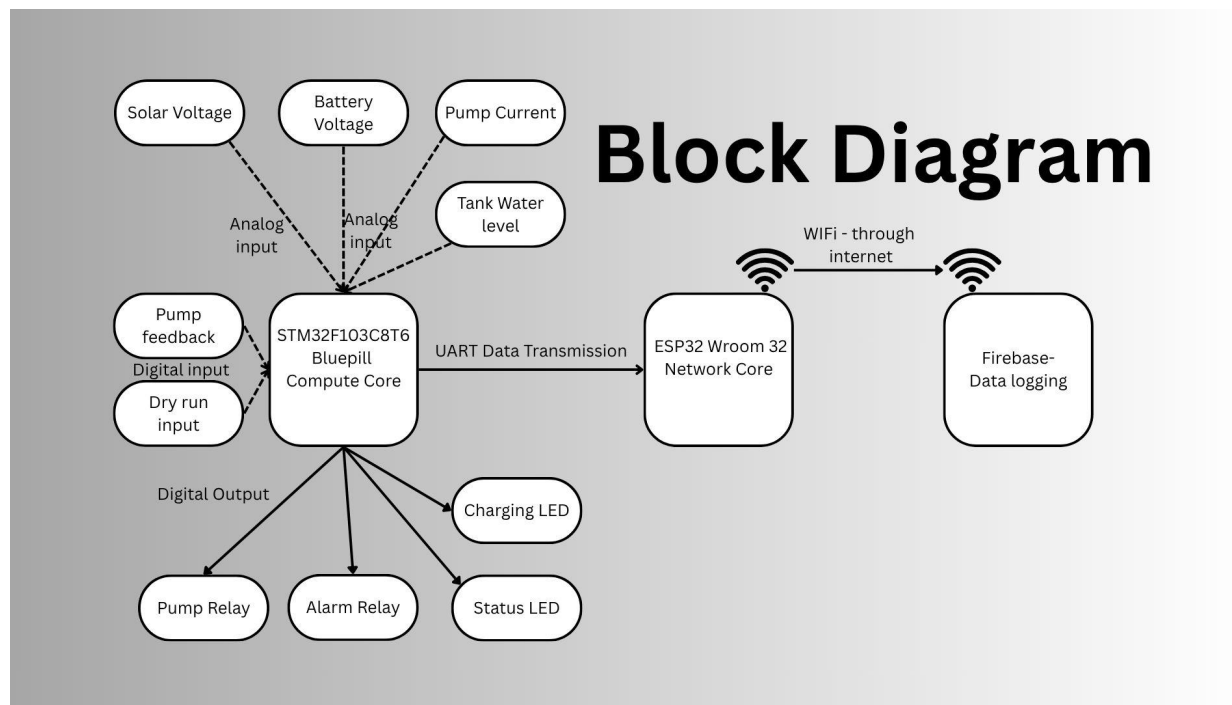
- Hall Sensor for Pump Feedback
- IR Sensor for Dry Run Detection

Outputs

- Pump LED
- Charging LED
- Alarm LED
- Buzzer
- Status LED

In this prototype V1.0 using LEDs instead of Relays.

4. Block Diagram



5. GPIO Allocation Table

STM32 Pin	Function
PA0	Solar Voltage ADC
PA1	Battery Voltage ADC
PA2	Pump Current ADC
PA3	Tank Level ADC - Water Level Sensor
PA4	Pump Feedback Input - Hall Sensor
PA5	Dry Run Input - IR Sensor
PB0	Pump LED
PB1	Charging LED
PB10	Alarm LED
PB11	Buzzer
PC13	Status LED
PA9	UART TX
PA10	UART RX - not used in this prototype V1.0

6. ADC Measurement

STM32 uses a 12-bit ADC for analog measurements.

ADC Resolution:

4096 Levels (0–4095)

Reference Voltage:

3.3V

Voltage Conversion Formula:

$$\text{Voltage} = (\text{ADC Value} \times 3.3) / 4095$$

The ADC channels monitor:

- Solar Panel Voltage
- Battery Voltage
- Pump Current
- Tank Level

Tank level is converted into percentage for monitoring and decision-making.

```
solar_adc   = Read_ADC_Channel(ADC_CHANNEL_0);
battery_adc = Read_ADC_Channel(ADC_CHANNEL_1);
current_adc = Read_ADC_Channel(ADC_CHANNEL_2);
tank_adc    = Read_ADC_Channel(ADC_CHANNEL_3);

solar_voltage = (solar_adc * 3.3f) / 4095.0f;

battery_voltage = (battery_adc * 3.3f) / 4095.0f;

pump_current    = (current_adc * 10.0f) / 4095.0f;    // Simulate 0-10A

tank_level      = (tank_adc * 100.0f) / 4095.0f;      // 0-100%
// Charging Logic
```

From the STM32Cube IDE main.c

7. Pump Control Logic

The controller automatically starts the pump when water is required.

Pump Start Conditions

The pump is allowed to run only if:

- Tank level is below 20%
- Battery voltage is above 2.3V
- Dry-run condition is not present
- Pump feedback is available

When all conditions are satisfied:

- Pump LED turns ON
- Pump status becomes active

Pump Stop Conditions

The pump is stopped when:

- Tank level exceeds 80%
- Battery voltage becomes low
- Dry-run fault occurs
- Pump feedback is lost

```
if(tank_level < TANK_LOW_LEVEL)
{
    if(battery_voltage > BATTERY_MIN_VOLTAGE)
    {
        if(dryRun == 1)
        {
            // Start Pump
            LED_ON(GPIOB, GPIO_PIN_0);
            LED_ON(GPIOC,GPIO_PIN_13);
            pumpStatus = 1;
            //HAL_Delay(2000);
            if(pumpFeedback)
            {
                // Normal Operation
                LED_OFF(GPIOB, GPIO_PIN_10);
                BUZZER_OFF();
                pumpStatus = 1;
                LED_ON(GPIOC, GPIO_PIN_13);
            }
        }
    }
}
```

```

else
{
    // Pump Feedback Fault
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_ON(GPIOB, GPIO_PIN_10);
    BUZZER_ON();
    LED_OFF(GPIOC,GPIO_PIN_13);
}
}
else
{
    // Dry Run Fault
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_ON(GPIOB, GPIO_PIN_10);
    BUZZER_ON();
    LED_OFF(GPIOC,GPIO_PIN_13);
}
}
else
{
    // Low Battery Fault
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_ON(GPIOB, GPIO_PIN_10);
    BUZZER_ON();
    LED_OFF(GPIOC,GPIO_PIN_13);
}
}
else if(tank_level >= TANK_HIGH_LEVEL)
{
    // Tank Full
    LED_OFF(GPIOB, GPIO_PIN_0);
    pumpStatus = 0;
    LED_OFF(GPIOB, GPIO_PIN_10);
    BUZZER_OFF();
    LED_ON(GPIOC,GPIO_PIN_13);
}
}

```

8. Pump Feedback Monitoring

A Hall sensor is used to simulate pump feedback.

After the pump is enabled, the controller verifies pump feedback.

If pump feedback is missing:

- Pump operation is stopped
- Alarm LED is activated
- Buzzer is activated
- Fault condition is generated

This ensures the pump is operating correctly.

```
if(dryRun == 1)
{
    // Start Pump

    LED_ON(GPIOB, GPIO_PIN_0);
    LED_ON(GPIOC, GPIO_PIN_13);
    pumpStatus = 1;
    //HAL_Delay(2000);

    if(pumpFeedback)
    {
        // Normal Operation

        LED_OFF(GPIOB, GPIO_PIN_10);

        BUZZER_OFF();
        pumpStatus = 1;
        LED_ON(GPIOC, GPIO_PIN_13);
    }
}
```

9. Solar Charging Indication

Solar voltage is monitored continuously.

When solar voltage exceeds 2.5V:

- Charging LED is activated

This indicates that sufficient solar power is available for charging operation.

```
187 // Charging Logic
188 if(solar_voltage > SOLAR_CHARGE_VOLTAGE && battery_voltage < 3.2)
189 {
190     LED_ON(GPIOB, GPIO_PIN_1); // Charging LED
191     LED_ON(GPIOC,GPIO_PIN_13);
192     chargingStatus = 1;
193 }
194 else
195 {
196     LED_OFF(GPIOB, GPIO_PIN_1);
197     LED_ON(GPIOC,GPIO_PIN_13);
198     chargingStatus = 0;
199 }
200
```

10. UART Communication

STM32 transmits system parameters to ESP32 every 3 seconds.

Communication Settings:

- Baud Rate: 115200 bps
- Data Bits: 8
- Stop Bits: 1
- Parity: None

Example UART Data:

SV=2.45,BV=2.80,PC=1.20,TL=45.0,PF=1,DR=0,PS=1,CS=0

The ESP32 parses the received data and updates the cloud database.

Firestore

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Project Overview
Settings

Project shortcuts

- Realtime Database
- Phone Verification **NEW**
- Authentication

Product categories

- Databases and storage
- Security
- AI services
- Hosting and serverless
- DevOps and engagem...
- Analytics

Spark
No cost (\$0/month)

Upgrade **NEW**

Solar Pump Monitoring Control

Realtime Database

Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing [Configure App Check](#)

<https://solar-pump-monitoring-control-default-rtdb.asia-southeast1.firebaseio.com>

```
https://solar-pump-monitoring-control-default-rtdb.asia-southeast1.firebaseio.com/  
  batteryVoltage: 3.29  
  chargingStatus: 0  
  dryRunStatus: 1  
  pumpCurrent: 2.01  
  pumpFeedback: 1  
  pumpStatus: 1  
  solarVoltage: 0  
  tankLevel: 0
```

COM4 - PuTTY

```
SV=2.40,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.51,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.59,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.60,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.60,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.60,BV=3.29,PC=2.25,TL=25.3,PF=1,DR=1,PS=0,CS=0  
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SV=2.86,BV=3.29,PC=2.25,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=2.97,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=3.03,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=3.07,BV=3.29,PC=2.24,TL=25.2,PF=1,DR=1,PS=0,CS=0  
SV=3.11,BV=3.29,PC=2.25,TL=25.2,PF=1,DR=1,PS=0,CS=0  
SV=3.14,BV=3.29,PC=2.25,TL=25.2,PF=1,DR=1,PS=0,CS=0  
SV=3.18,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
SV=3.21,BV=3.29,PC=2.24,TL=25.3,PF=1,DR=1,PS=0,CS=0  
S
```

11. Firebase Cloud Monitoring

Firebase Realtime Database is used for remote monitoring.

Uploaded Parameters:

- Solar Voltage
- Battery Voltage
- Pump Current
- Tank Level
- Pump Feedback
- Dry Run Status
- Pump Status
- Charging Status

The cloud database allows real-time monitoring of the solar pump system from any internet-connected device.

12. Project Outcome

The developed system successfully demonstrates:

- Solar pump automation
- Battery protection
- Dry-run protection
- Pump feedback monitoring
- Real-time parameter acquisition
- UART communication between STM32 and ESP32
- Cloud-based monitoring using Firebase

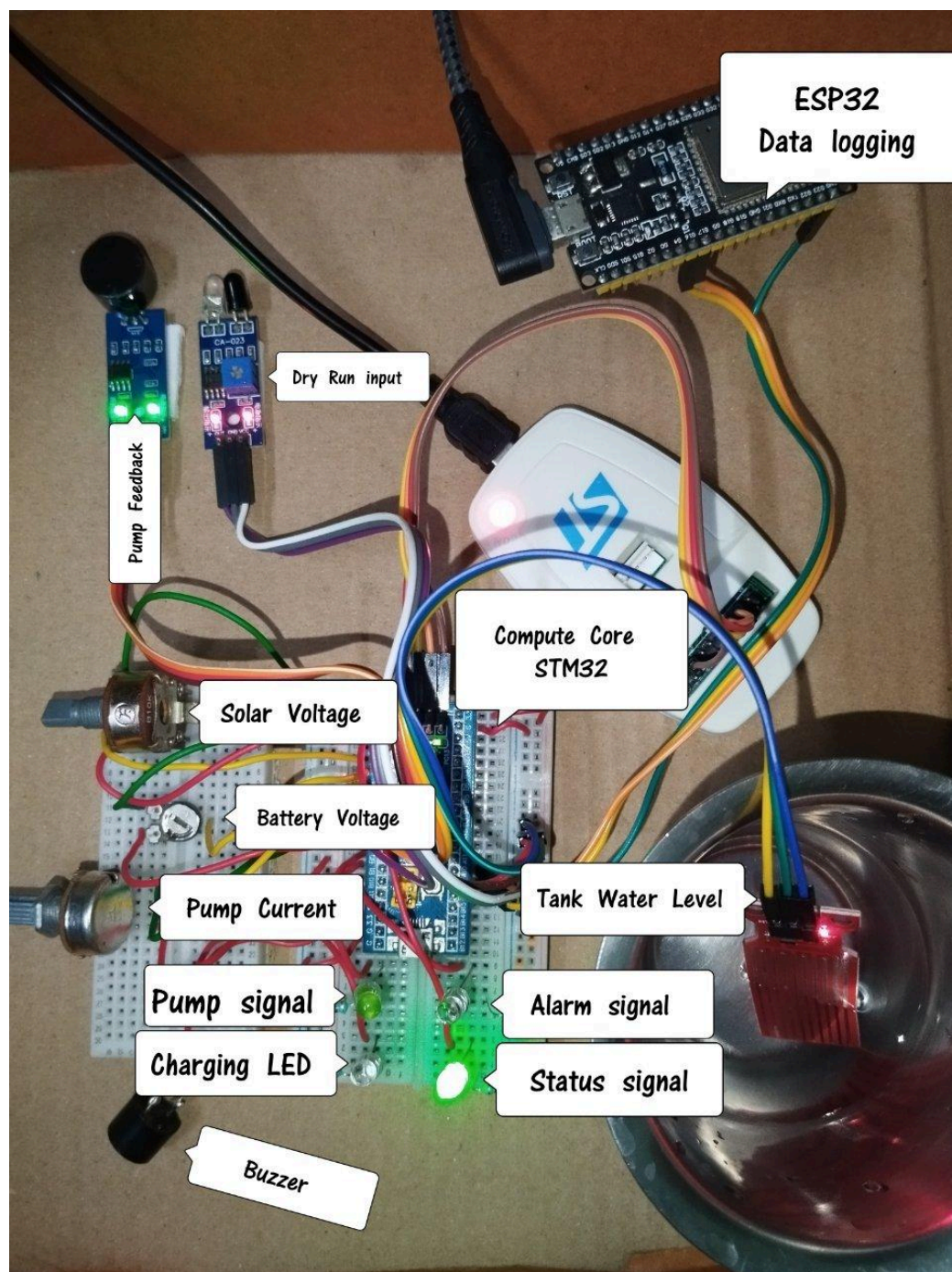
13. Project Outcome

Working Prototype Videos:  Olive lot Assignment task 9 videos

Source Codes: Github link:

<https://github.com/arivenkkataram-asj/Solar-Pump-Monitoring-Controller.git>

11. Working Prototype Images:



ESP32
Data logging

Dry Run input

Pump Feedback

Compute Core
STM32

Solar Voltage

Battery Voltage

Pump Current

Tank Water Level

Pump signal

Alarm signal

Charging LED

Status signal

Buzzer

